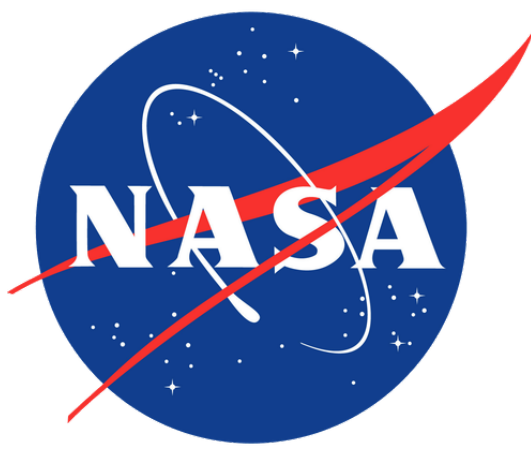


Training and Benchmarking AI/ML Models Using Space Biology Transcriptomic Data

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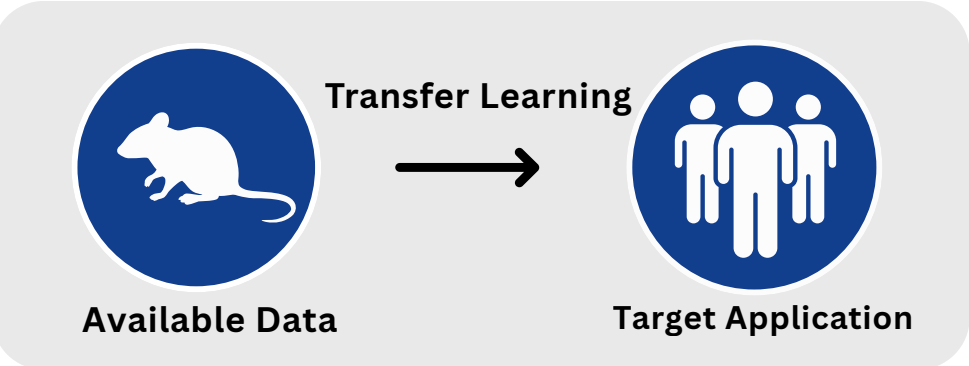
Overview

Problems and Context

- Space missions are growing longer, posing more risks to astronauts
- No existing foundational models for interpreting spaceflight experiments and downstream tasks (see examples below)
- Spaceflight experiments are expensive and human data is rare and regulated

Our Approach

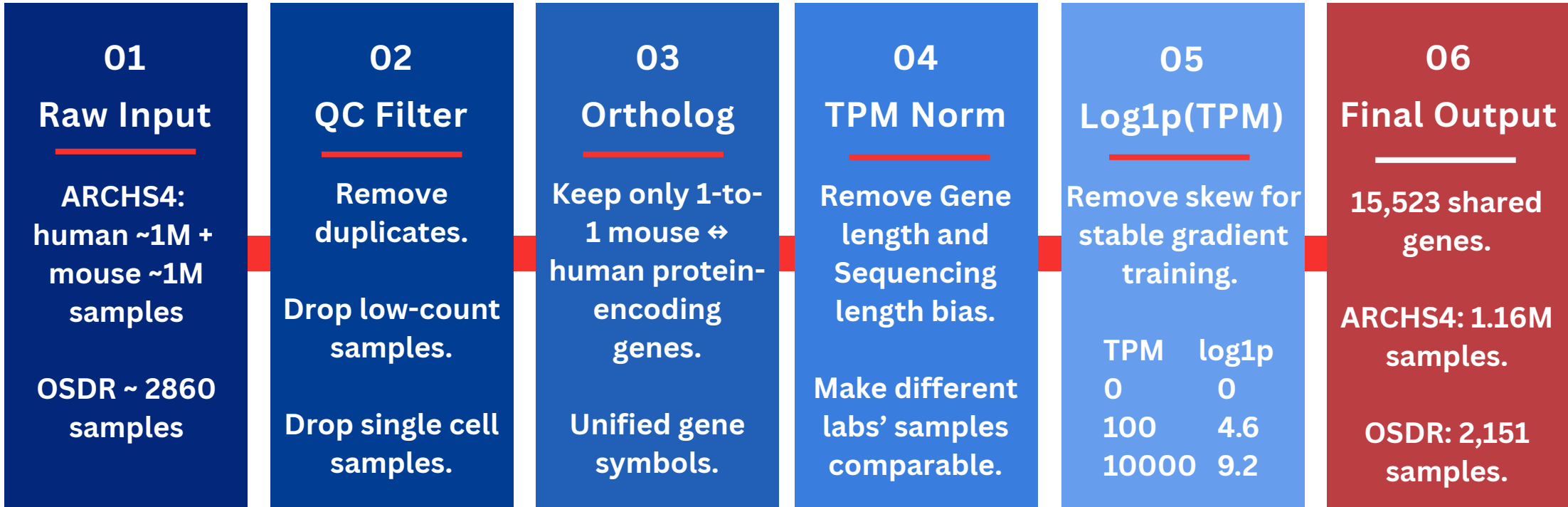
Train a foundational model on 20k human and mouse bulk RNA-seq samples to be fine-tuned for space biology downstream tasks. To combat the problem of having only space mouse data, we experimented with 3 main training datasets: mouse data only, human data only, and half of each



Downstream Task Examples:

- Simulate spaceflight's effects on humans
- Predict missing values in spaceflight data
- Early detection of radiation

Data + Preprocessing Pipeline



Sample retention after preprocessing



Model Architecture - Pretraining

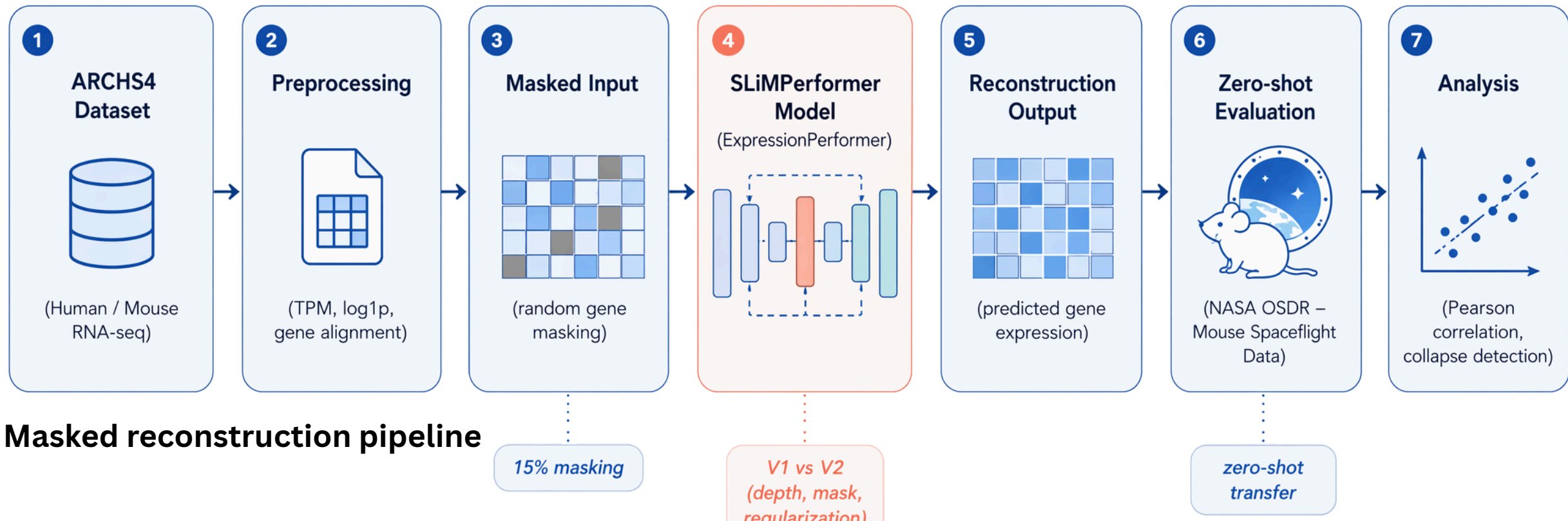


BulkFormer

The state-of-the-art model we used as a baseline was the BulkFormer, a model trained on ground human bulk RNA-seq data; no understanding of spaceflight data.

Why bulk RNA-seq?

It comes down to cost, coverage, and compatibility. Cheaper than single-cell, give a fuller context of the sample, and fits NASA's sampling type.



Masked reconstruction pipeline

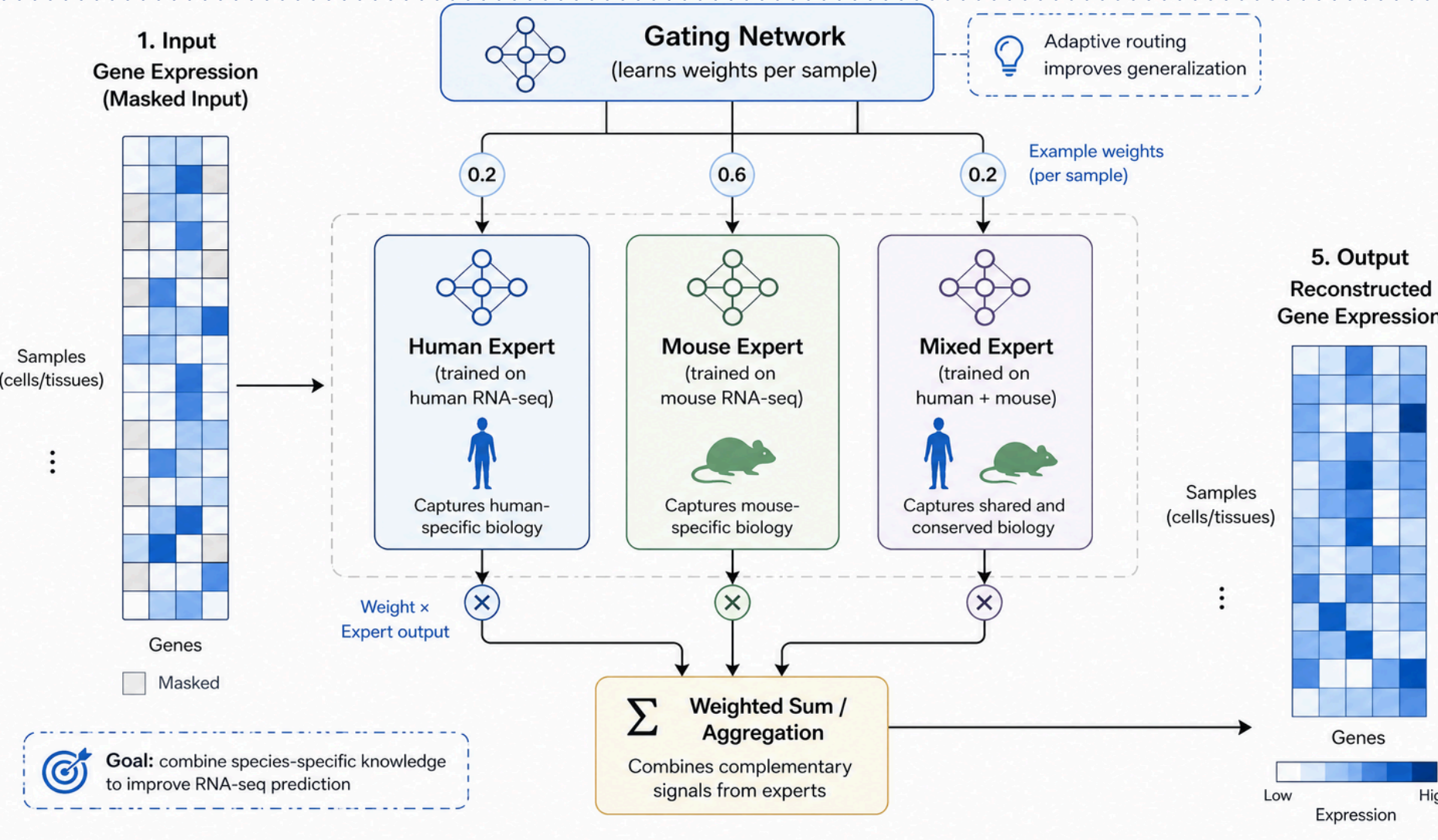
Train (ARCHS4)	Zero-shot Test (OSDR, mouse)
Mouse (5k)	0.78
Mixed (5k)	0.76
Human (5k)	0.69
Human (20k)	0.81

Model architecture overview

Models are trained with masked reconstruction on ARCHS4 and evaluated zero-shot on OSDR (mouse) to study generalization.

What limits generalization and how to fix it?

At small scale, models learn a shortcut by predicting the gene mean, limiting generalization despite capturing some biological signal. Scaling data breaks this shortcut and enables structured, sample-specific predictions, leading to large performance gains. Architectural changes help less, indicating that scale is the primary driver of generalization and that stronger base models are needed before applying methods like mixture-of-experts.



Mixture-of-Experts (MoE)

We also explore a Mixture-of-Experts (MoE) approach to combine species-specific models via adaptive gating. While this is conceptually promising, oracle analysis shows near-zero headroom, as all experts exhibit the same collapse behavior. This suggests that MoE is currently limited by expert quality rather than routing, and future work should focus on scaling and improving base models before revisiting expert combination.

Weights and Biases

Tool used to find best hyperparameter configs such as:

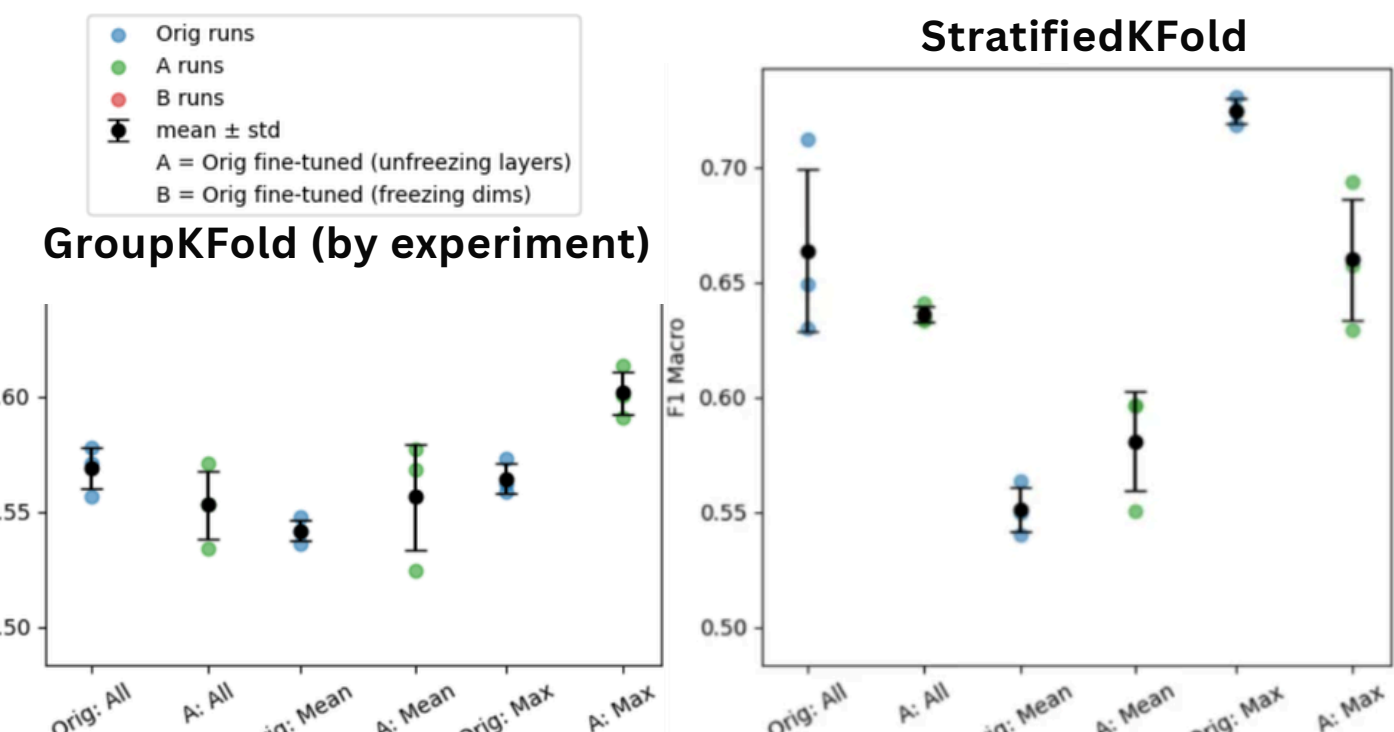
- batch_size
- learning_rate
- hidden_dim
- num_layers



Model Architecture - Finetuning

Task 1: binary classification of spaceflight / non-spaceflight

We fine-tuned a pretrained backbone (using half human, half mouse data) by gradually unfreezing layers per epoch. Other approaches were tested, but this method (Model A) performed best.



Evaluating various embeddings:

We trained an MLP on aggregated embeddings (mean/max/all) from the two models, and evaluated classification using F1 score with StratKFold & GroupKFold cross-validation. Max aggregation performed best, and the fine-tuned model seems to have learned more experiment-independent representations.

Visualizing embeddings:

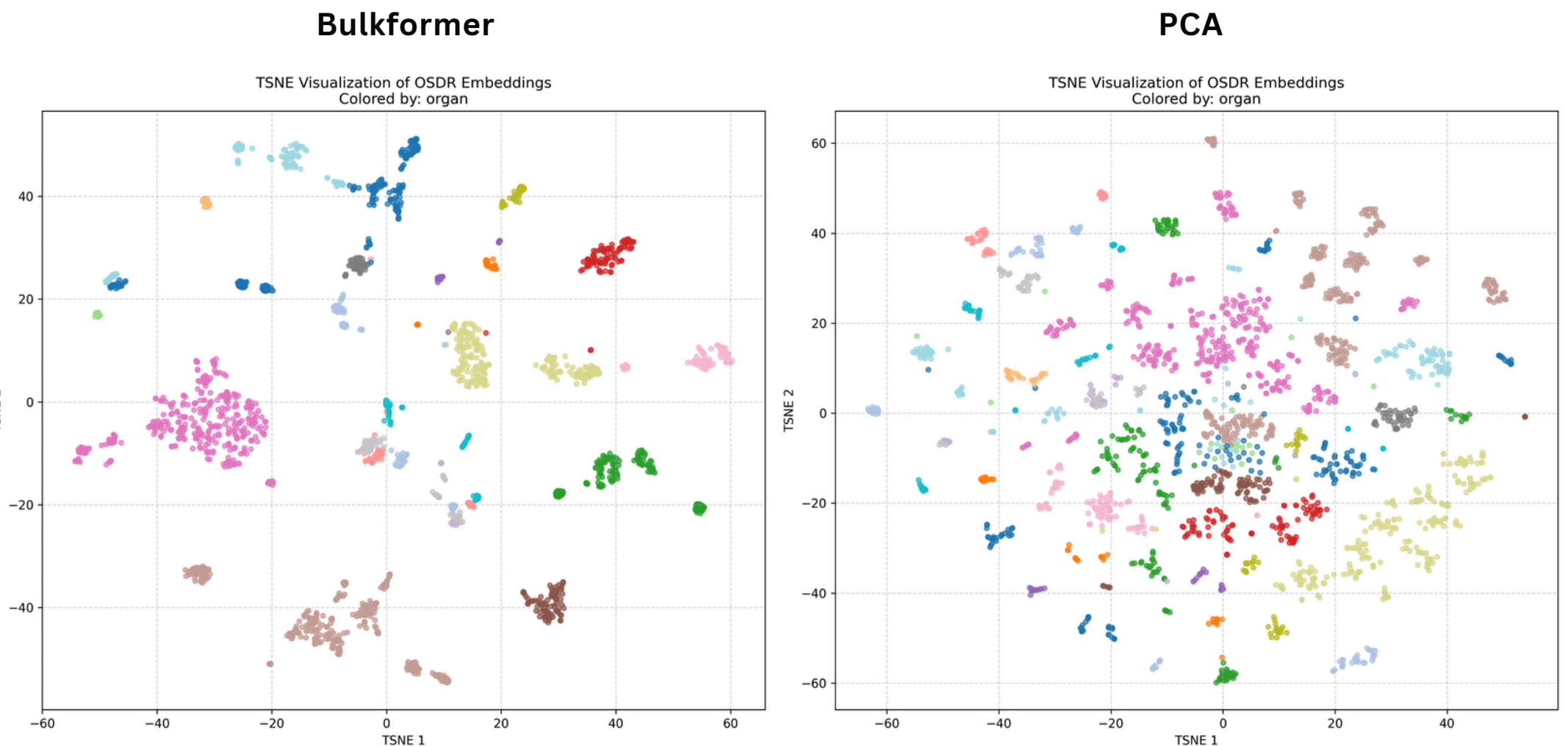
PCA of sample embeddings shows no clear separation between spaceflight and non-spaceflight, despite F1 > 0.6. Model A exhibits greater intra-experiment spread, suggesting improved performance may stem from increased sample variability. Clustering patterns indicate tissue type may strongly influence classification, as every experiment's samples generally come from the same tissue type.

Task 2: classification of tissue type

Method	Accuracy
Bulkformer (our model)	80%
PCA (baseline)	44%

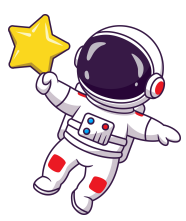
OSDR Tissue Type Classifier

We extracted embeddings for all of NASA's mouse samples using Bulkformer and PCA, and trained a logistic regression model to predict tissue type. Train/eval sets were selected such that there was no batch leakage. Our Bulkformer model convincingly outperforms the PCA baseline.



Future Steps

Future research could investigate scaling the aforementioned models with higher computational resources, retraining model architectures on human spaceflight data, and exploring other general bulk RNA-seq genomics models as choices for the backbone.



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